A Scalable Framework for Detecting and Identifying Novel Colloidal Crystals

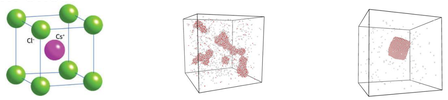
Steven Zhang[1], JP Marquardt[2], Philipp Hoellmer[2], Nicole Smina[2], Glen Hocky[2]

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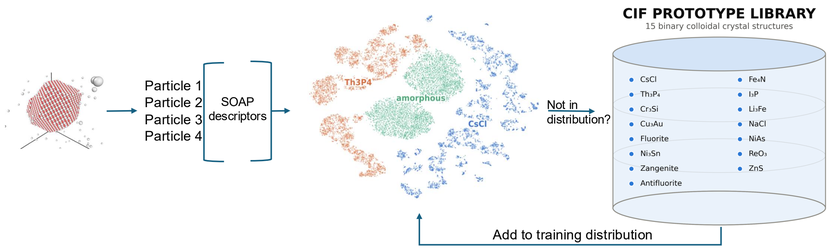
1. Yale University, 2. New York University

Motivation

Colloidal particles can self-assemble into many crystal structures based on size, attraction levels, and brush lengths [5]. Molecular dynamics simulations allow for large parameter sweeps to generate millions of particle configurations [3]. However, manual labeling and structure inspection do not scale. Closed-set classifiers force unseen structures into existing classes. A useful discovery system must both **classify known structures** and **abstain on unknown ones**.

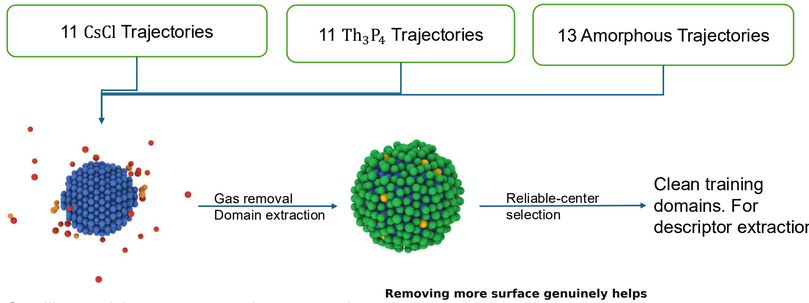


Developing an OOD Detector and Matcher Pipeline

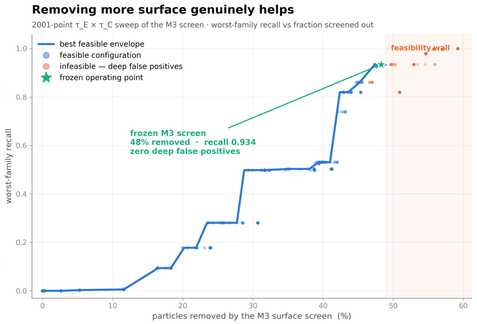


The pipeline first detects structural domains by removing gas-like particles, separating connected components, and comparing species-aware SOAP fingerprints with the known training distribution using Mahalanobis distance. Domains flagged as unfamiliar are then matched against a library of relaxed CIF prototypes. After a structure is independently and manually verified, it is used to expand the training atlas, allowing future simulations to recognize a broader range of crystal structures.

Building a Training Atlas by Flagging Noisy Particles



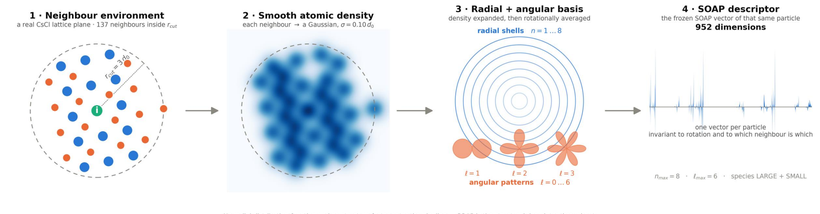
Gas-like particles are removed, connected domains are extracted, and a phase-blind surface rule selects reliable crystal centers while preserving the complete component as neighborhood context. Gas extraction uses coordination and the mean distance to the four nearest neighbors. A depth-first-search based graph then separates connected domains. Surface filtering improves out-of-distribution detection up to an optimal removal level, after which useful structural information begins to be lost (seen in the figure on the right).



From Local Particle Environments to Domain-level Novelty

A. Encode each local environment with SOAP (smooth overlap of atomic positions)

SOAP describes each particle's local atomic environment in a way that is invariant to translation, rotation, and permutation of identical particles. For every retained center particle, neighboring LARGE and SMALL particles are represented as separate smooth density fields [1].



$$p_i^a(r) = \sum_{j \in \alpha} \exp\left[\frac{-|r - r_{ij}|^2}{2\sigma^2}\right] f_{cut}(r_{ij})$$

Each neighbor is replaced by a Gaussian cloud. Summing the clouds produces a smooth, species-resolved density around center particle. The equation below shows the SOAP density expansion in radial and spherical-harmonic bases [2].

$$p_i^a(r) = \sum_{n,l,m} c_{nlm}^{i,a} g_n(r) Y_{lm}(\theta, \phi)$$

B. Construct a domain fingerprint

Each particle's 952-dimensional SOAP vector is normalized, standardized according to its center species, and compressed to 16 dimensions using a shared Principal Component Analysis (PCA). Large and small centered vectors are then averaged separately and concatenated into one 32-dimensional fingerprint representing the entire domain.

$$\begin{aligned}
1.\ \text{SOAP vector} &\Rightarrow\ x_i \in \mathbb{R}^{952} \\
2.\ \text{L2 normalize} &\Rightarrow\ \hat{x}_i = \frac{x_i}{\|x_i\|_2} \\
3.\ \text{Standardize} &\Rightarrow\ \tilde{x}_i = \frac{\hat{x}_i - \mu_{s_i}}{\sigma_{s_i}}, s_i \in \{L, S\} \\
4.\ \text{PCA} &\Rightarrow\ z_i = W\tilde{x}_i \\
5.\ \text{Average} &\Rightarrow\ \bar{Z}_L = \text{mean}(Z_L), \bar{Z}_S = \text{mean}(Z_S) \\
6.\ \text{Concatenate} &\Rightarrow\ h = \begin{bmatrix} \bar{Z}_L \\ \bar{Z}_S \end{bmatrix} \in \mathbb{R}^{32}
\end{aligned}$$

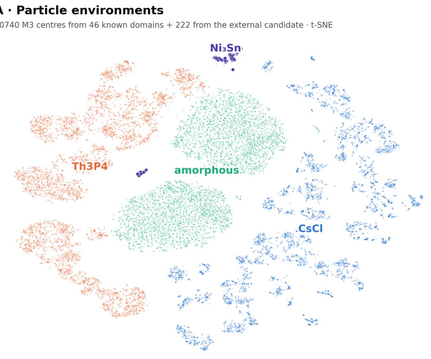
C. Detect novelty using Mahalanobis distance

The domain-level fingerprint is compared with the known training distribution using Mahalanobis distance, which accounts for both feature variance and correlations. The distance is normalized by a calibrated threshold (τ) to classify the domain as known-like or out-of-distribution [7, 8].

$$d_M(h) = \sqrt{(h-\mu)^T \sum_{LW}^{-1} (h-\mu)}$$

$$\Sigma = \begin{bmatrix} Var(h_1) & Cov(h_1, h_2) & \dots \\ Cov(h_2, h_1) & Var(h_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

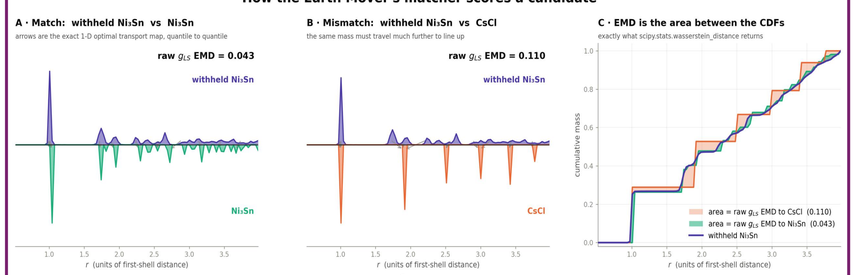
$$\rho = \frac{d_M}{\tau} \qquad \begin{aligned} p \le 1&: \text{KNOWN} \\ p > 1&: \ OOD \end{aligned}$$



t-SNE visualization only, where Ni_3Sn is a new sample [10]

Validating against a CIF Library

Novel domains are compared against a library of relaxed CIF files using species-resolved radial distribution functions $g_{LL}(r)$, $g_{LS}(r)$, $g_{SS}(r)$, $g_{total}(r)$, and the structure factor $S(k)$ [4, 6]. Candidate and reference curves are compared using Earth Mover's Distance (EMD), while a Bragg coherence check confirms long-range crystalline order [9].



1. $W_1(P, Q) = \int |F_P(x) - F_Q(x)| dx$

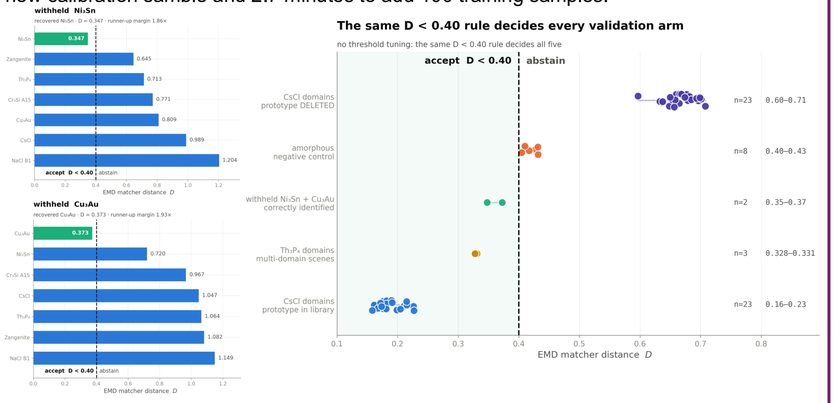
2. $S_f = \max_c \left(\frac{d_{f,c}}{\tau_{f,c}}\right)$, $S_f \le 1 \Rightarrow$ reference accepted

$* \ c \in \{species - resolved\ RDF, total\ RDF, S(k)\}$

Earth mover's distance is the area between the cumulative candidate and reference curves; smaller values indicate greater structural similarity **(1)**. Each distance is normalized by a family-specific tolerance. The worst channel determines whether the prototype f passes, preventing one good curve from hiding a poor match **(2)**.

Results and Conclusions

OOD Mahalanobis detection achieved TPR = 1.00 and AUROC = 1.00 for leave-one-family-out tests across CsCl, Th_3P_4, and amorphous domains. The prototype matcher correctly identified all CsCl and Th_3P_4 domains when CsCl and Th_3P_4 were present in the library, rejected all domains when the reference was removed, and flagged two examples and matched them correctly (figures below). The pipeline takes less than one second to add a new calibration sample and 2.7 minutes to add 100 training samples.



References

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